

JUMBOLINE OPTICAL NETWORKS

Turning Churn Data Into Retention Action

From raw customer data to a quantified, prioritized retention program, and a live dashboard to run it

7043

Customers Analyzed

26.5%

Overall Churn Rate

\$139K

Monthly Revenue at Risk

About Jumboline



Mission

To be the provider of choice for flexible, budget-friendly telecom and streaming bundles that keep families connected on their own terms.



Vision

To redefine regional connectivity through transparent pricing, zero lock-in contracts, and customer-first service.

The Business Challenge



New customer sign-ups
have stayed healthy and
consistent



Recurring revenue
growth has stalled
despite that acquisition



Jumboline's VP of Customer Retention commissioned an analysis, after a quarterly business review where acquisition metrics looked strong but net subscriber growth barely moved, a gap only a churn-specific, customer-level analysis could explain.

Objectives & Approach

What this analysis set out to do, and how

1

Build a Live Risk Framework

- ✓ Flag currently active customers before they churn, not just explain who already left.

2

Test Whether Levers Compound

- ✓ Determine if retention tactics reinforce or duplicate each other, so spend isn't wasted.

3

Package It Into a Self-Serve Tool

- ✓ A dashboard the retention team can operate independently, without rerunning the analysis.



Python for cleaning and exploratory analysis



PostgreSQL for validated business logic



Power BI for a live, interactive decision-support tool

The Scale Of The Problem

One in four customers leaves, and it is already showing up on the revenue line

26.5%

- Overall churn rate across 7,043 customers

\$139,131

- Monthly recurring revenue tied to churned customers

30.5%

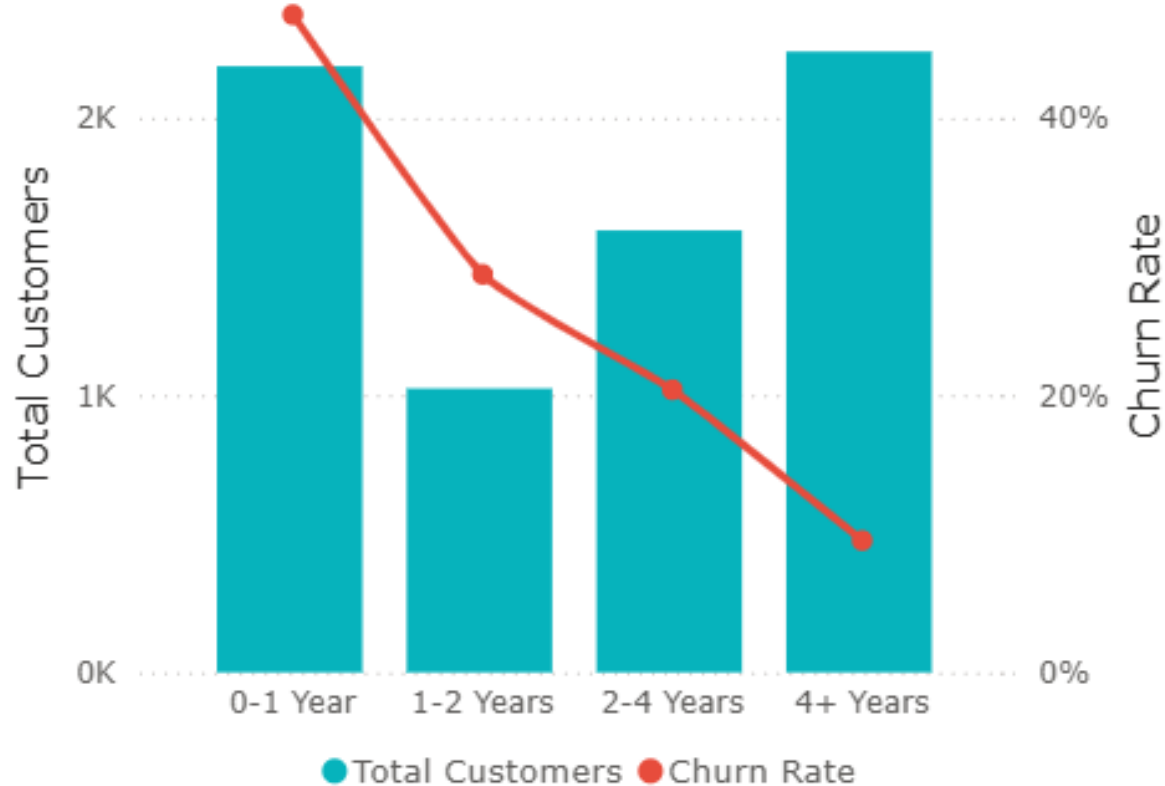
- Share of total monthly revenue already lost to churn

This is not a marginal problem.

A quarter of the customer base leaving, and nearly a third of monthly revenue already gone with them, is a scale that justifies a dedicated, funded retention program.

When We Lose Customers

Churn is a first-year problem, not a steady drip



47.4%

Churn rate in a customer's first year alone

5x

Higher churn in year 1 vs. the 4+ year cohort

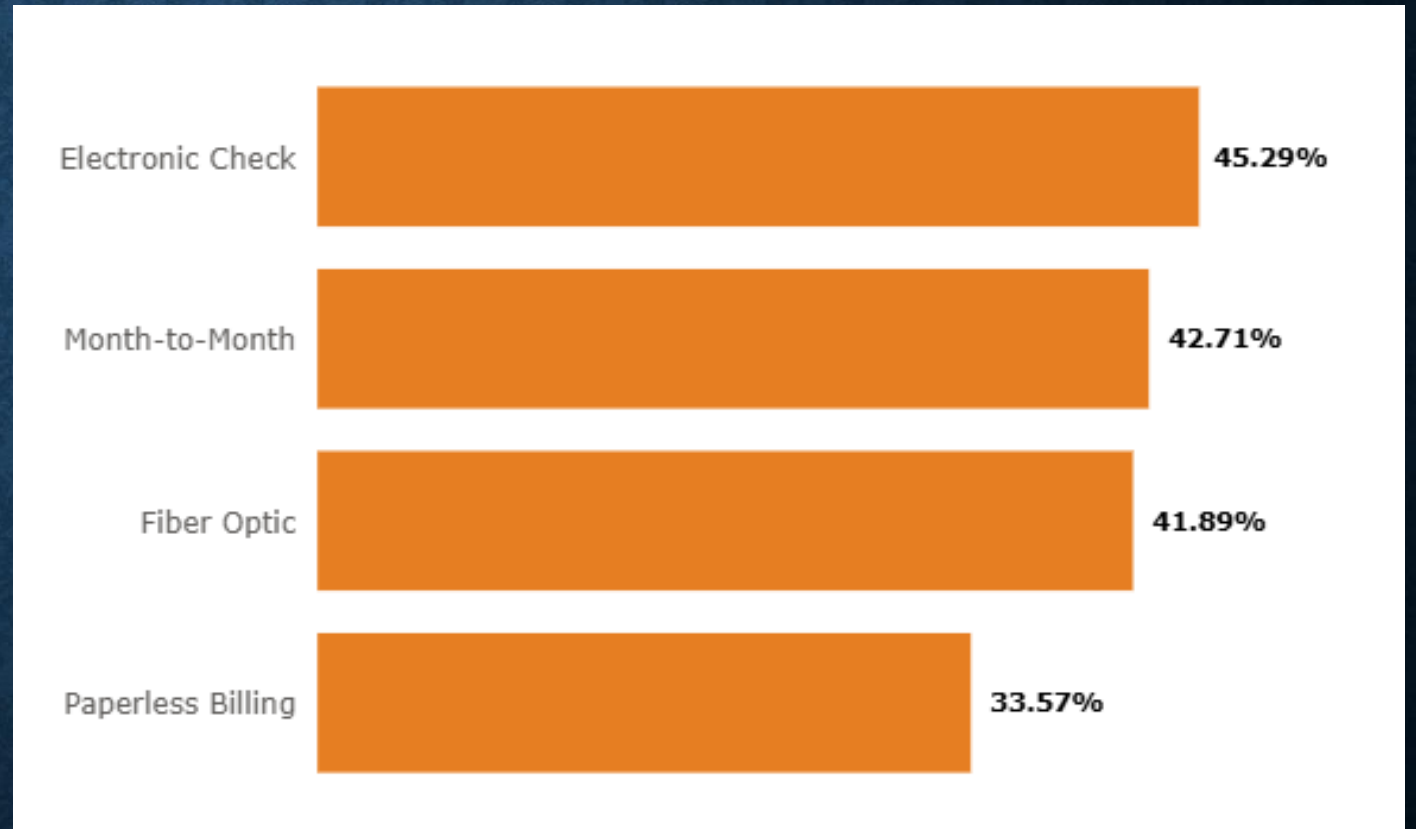


Retention effort concentrated in the first 12 months addresses the single largest share of churn.

The Four Warning Signs



Four factors
repeatedly
separate high-risk
customers from
stable ones



Risk Profile

<=12 months customers on Month-to-month contract, fiber optic internet, electronic check payment

2-, 3-, and 4-way intersection:

	Fiber Optic	DSL	No Internet
Month to Month	54.61%	32.22%	18.29%
One Year	19.29%	9.30%	2.47%
Two Year	7.23%	1.91%	0.78%

	Fiber Optic +			
	Electronic Check	Mailed Check	Bank Transfer (Automatic)	Credit Card (Automatic)
Month to Month	60.37%	50.75%	45.57%	41.64%
One Year	26.02%	19.44%	14.65%	15.33%
Two Year	9.78%	4.76%	9.26%	3.90%

	Fiber Optic + Electronic Check +			
	0-1 Year	1-2 Years	2-4 Years	4+ Years
Month to Month	71.16%	58.82%	48.95%	37.04%
One Year	25.00%	25.00%	28.81%	24.81%
Two Year	--	--	20.00%	8.64%

182 Critical Risk Customers

Matching all the four factors

782 Retention Targets

Matching 3+ factors

19.18% MMR Exposure

Share of active revenue held by retention targets

The Business Case for Action

Modeled savings from reducing churn in the highest-risk segment (month-to-month, fiber optic)

Priority Lens

High Churn/High Value	High Churn/Low Value
Low Churn/High Value	Low Churn/Low Value

- ❖ A high churn rate in a tiny segment may have limited aggregate impact.
- ❖ A large segment with moderately elevated churn can create greater recurring exposure.
- ❖ Retention should target where risk (churn rate), scale (customer volume), and recurring revenue overlap.

Prioritize:

Tenure	Early lifecycle customers
Contract	Month-to-month
Internet	Fiber optic
Payment	Electronic check
Value	Higher monthly charges

Scenario:

Revenue Preservation Per % Churn Reduction in Month to Month + Fiber

\$10,048

Saved MMR By 10% Churn Reduction

\$20,096

Saved MMR By 20% Churn Reduction

Two Levers That Compound

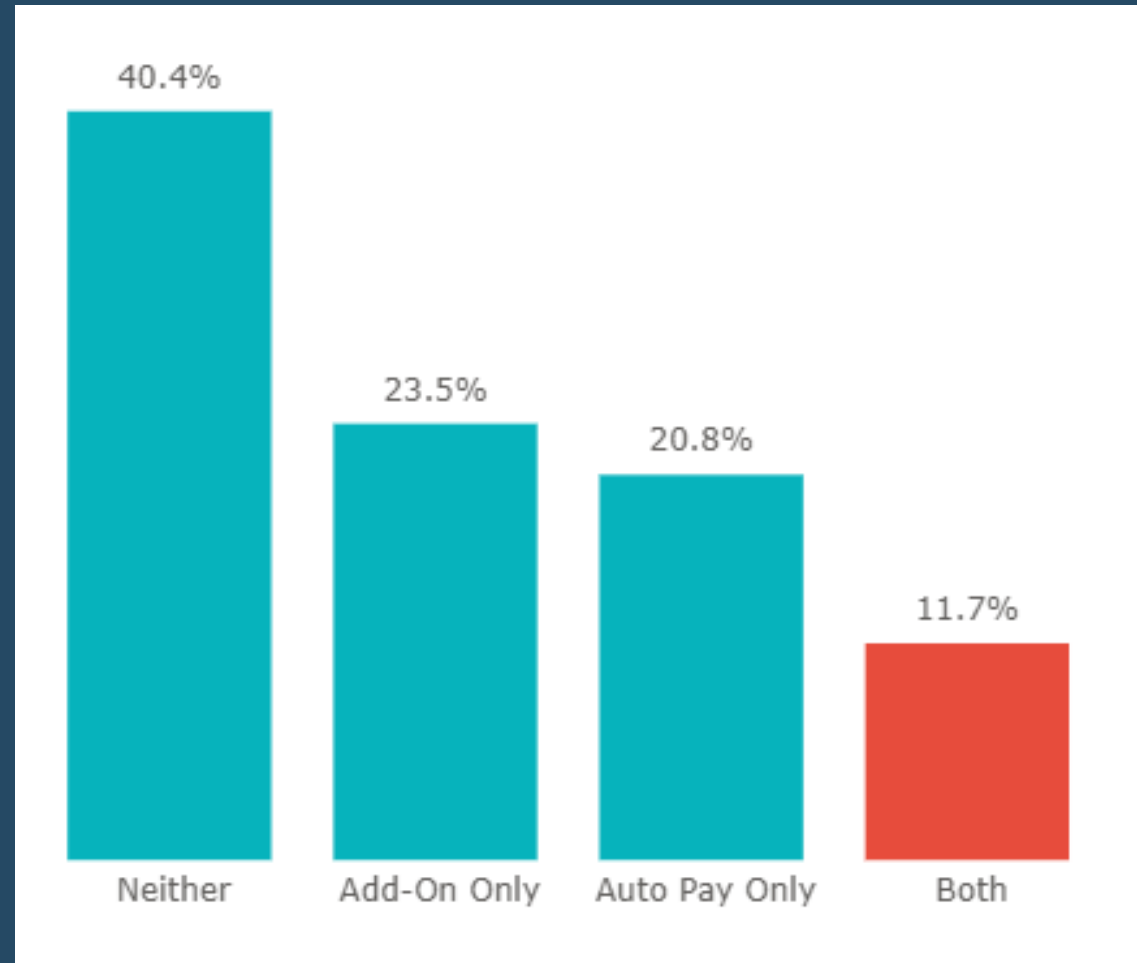
Protective add-ons and automatic payment each help, together they help far more

- ❖ Online Security and Tech Support are associated with lower churn.
- ❖ Automated payment methods are associated with lower churn than electronic check.
- ❖ The combination has the lowest observed churn among the tested lever combinations.

$$\text{Relative Change} = \frac{11.7\% - 40.4\%}{40.4\%} \times 100 = -71\%$$

- ✓ Customers with "Both add-ons and auto-pay" have a 71% lower churn rate compared to "Neither Lever" customers.

Recommendation: pursue the add-on cross-sell and the payment migration together, not as separate initiatives.



Recommended Retention Strategy



Strengthen Early Lifecycle Retention

- Concentrate onboarding and proactive outreach in months 0-12, where churn risk peaks



Reduce Month-to-month Exposure

- Incentivize migration to one- or two-year contracts; churn drops from 42.7% to 11.3% and 2.8%, respectively



Audit The Fiber Optic Experience

- Investigate pricing, reliability, and service quality; fiber churns at double the DSL rate at every contract length



Migrate Electronic Check Customers

- Offer incentives to move to automatic bank transfer or credit card; the highest risk payment method by a wide margin

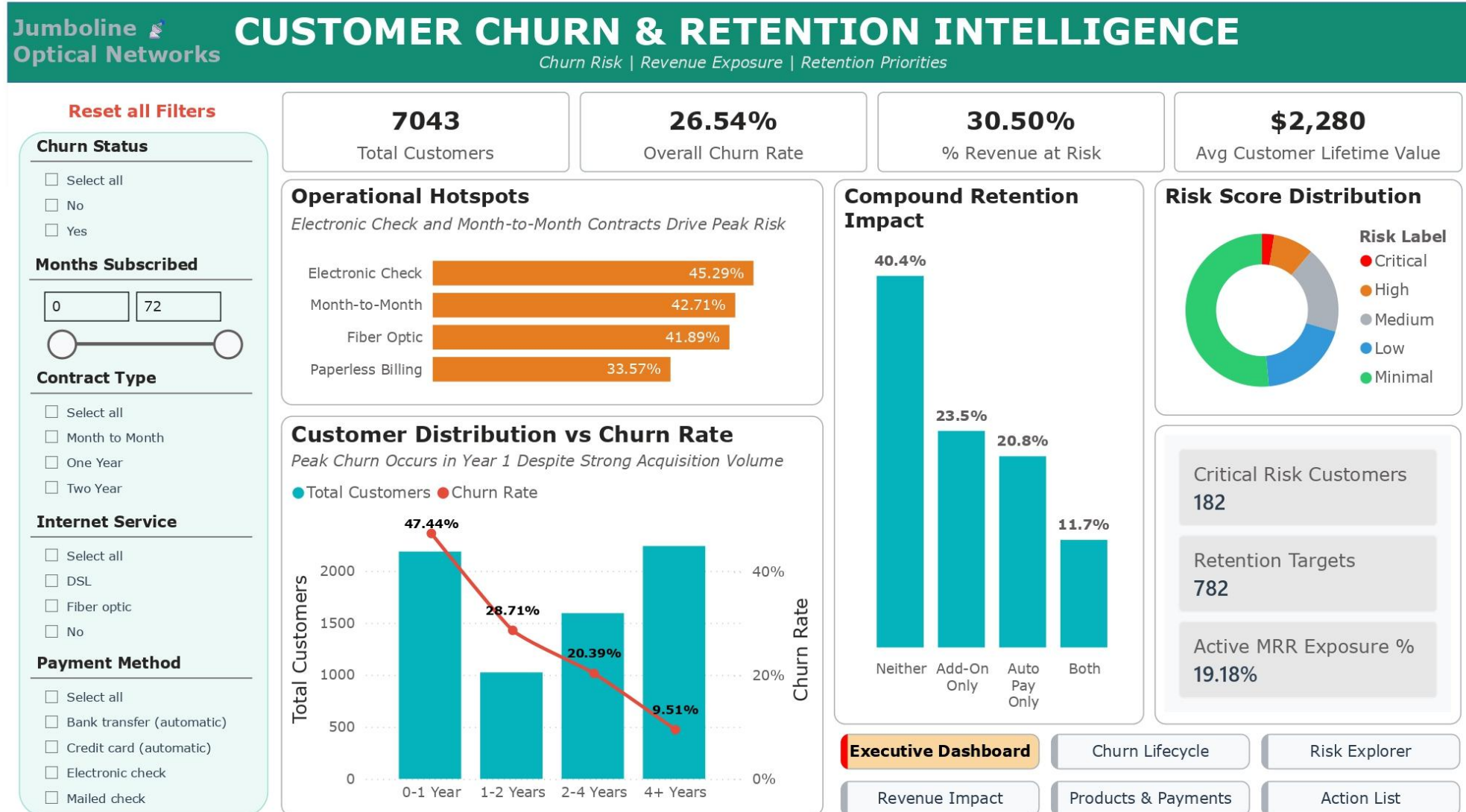


Bundle Add-ons With Auto-pay

- Combine the protective add-on upsell with payment migration; together, they cut churn to 11.7%. Each alone cuts less

A Live Decision-Support Tool

Six interactive pages, connected directly to the underlying database, refreshed on demand



Conclusion

Churn is not explained by any single attribute. It emerges from the intersection of contract flexibility, payment friction, and how early in the relationship a customer sits.

- ❖ Long-tenured customers and those on multi-year contracts are consistently the most stable.
- ❖ Newer, month-to-month, electronic-check customers carry the highest and most predictable risk.
- ❖ Risk is not fixed: retention levers compound, which means it's something Jumboline can actively influence, not just forecast.
- ❖ Even a low-risk profile isn't fully immune, fiber's elevated churn persists even under long contracts.

